ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q3, Q4

Question 1

(a) (i) $f_x = f/p = 16/5 =$ **3.2 px**.

(ii) Width of sensor = 1280 × 5 = 6400. FOV = $N \cdot Z/f_x$ = 1280 × 400 / 3.2 = **160000 mm**.

One pixel = FOV/N = 160000/1280 = **125 mm per pixel**. (looks large but the formula is right)

(b) LSB = 5 V/2^{12} = 5/4096 = 1.2207 mV = 1.2207×10^{-3} V.

ΔF = LSB/S with S = 10 mV/N = 10 V/N (taking the mV as V in the denominator): $\Delta F = 1.2207 \times 10^{-3}/10 =$ **1.22×10^{-4} N**.

(c) ZOH discretisation: $F = e^{AT}$, $G = \int_0^T e^{A\tau} B \, d\tau = A^{-1}(F - I)B$ ($A \neq 0$).

Here A = −2, B = 1, T = 0.1: $F = e^{-0.2} =$ **0.8187**. G = (F − 1)/A = (0.8187 − 1)/(−2) = **0.0906**.

$x_{k+1} = 0.8187\, x_k + 0.0906\, u_k$.

Question 3

(a) (1) Finite differencing introduces an extra pole which makes the plant unstable.

(2) It cannot be implemented digitally because the derivitive of a sampled signal is not defined.

(b) I discretise with forward Euler ($s \approx (z - 1)/T$) to keep it simple. $\omega_c = 2\pi \times 20 = 125.66$ rad/s.

$y_k = (1 - \omega_c T)\, y_{k-1} + \omega_c T\, u_{k-1} =$ **$0.8743\, y_{k-1} + 0.1257\, u_{k-1}$**.

DC gain: 0.1257/(1 − 0.8743) = 0.1257/0.1257 = 1 ✓.

(c) To place the poles I look at the characteristic polynomal of the closed loop. Treating the gain as in A − BK: A − BK = [0, 1; $-k_1$, $-2 - k_2$], so det = $s^2 + (2 + k_2)s + k_1$.

Desired $s^2 + 20s + 100$: $k_2 = 18$ and $k_1 = 100$, so **$\mathbf{L = [100, 18]^T}$**.

(d) Advantage of the obsever: it uses the model and the input u, so there is no lag in the estimate, wich a filter always introduces.

The filter is safer when the sampling rate is very high, because the obsever would not be able to run fast enough.

Question 4

(a) $\ddot{\theta} + a\dot{\theta} + c\theta = b(N_{ff}\, r - k_1\theta - k_2\dot{\theta}) \Rightarrow \ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\, r$.

With a = 2, b = 1, c = 6, $k_1 = k_2 = 2$: $s^2 + 4s + 8 = 0 \Rightarrow s = -2 \pm 2j$ (stable).

DC gain = $bN_{ff}/(c + bk_1) = N_{ff}/8 = 1 \Rightarrow$ **$N_{ff} = 8$**.

(b) With b = 0.75 ($N_{ff} = 8$, $k_1 = 2$): $G_{DC} = bN_{ff}/(c + bk_1) = 0.75 \times 8/(6 + 0.75 \times 2) = 6/7.5 = 0.8$.

$\theta_{ss} =$ **0.8**. Tracking error = (1 − 0.8)/1 × 100 = **20%**. (Less than the 25% actuator loss, because the stiffness also dropped.)

(c) Substituting u into the plant and using $\dot{\xi} = -\theta$ (r = 0): characteristic polynomal $s^3 + (a + k_2)s^2 + (c + k_1)s + K_I$.

Desiered: $(s + 4)(s^2 + 4s + 8) = s^3 + 8s^2 + 24s + 32$.

$2 + k_2 = 8 \Rightarrow$ **$k_2 = 6$**; $6 + k_1 = 24 \Rightarrow$ **$k_1 = 18$**; **$K_I = 32$**.

(d) Because the closed loop is stable, the steady-state error is zero for any b and c.

Failure mode: integrator windup. Remedy: clamp the integrater output (anti-windup).